

Multi-band passive radar imaging using satellite illumination

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Abstract—Over the last years the interest has grown over imaging capabilities of passive radars, [1] [2]. In particular, the MAPIS (Multi-channel Passive ISAR for military application) project funded by EDA (European defence agency) has shown the potential of ISAR imaging using passive radars based on different illuminators of opportunity, [3]. The results of MAPIS ([4]) demonstrated also how the obtained ISAR images can be effectively exploited for target classification and recognition [5] despite the limited range resolution. The main limitation of DVB-S-based radar imaging resides in the need for cueing the system to look in the direction where the target is. This is due to the limited power budget available in the DVB-S case and in the subsequent need for highly directive pencil-beam antennas. In this paper, the idea of passive radar imaging is further developed by proposing a multi-band passive radar system. In the proposed approach, target detection and imaging are considered two different tasks to be performed by two separate sub-systems: namely a cueing sub-system and an imaging one. The cueing sub-system is devoted to target detection, localisation, and mostly important it triggers and cues the subsequent DVB-S sub-system for target imaging. Different potential cueing sub-systems are investigated like other passive radars (based on FM, DVB-T, or other satellite communication signals) or passive emitter tracking (PET). This paper presents the system concept, and it proposes a potential experimental setup based upon hardware available at Fraunhofer FHR.

I. INTRODUCTION

The expression passive coherent location (PCL) indicates a class of bistatic radar systems that do not send a dedicated electromagnetic signal, but instead they exploit electromagnetic signals emitted by other sources for other purposes, [6]. Such sources are usually referred to as “illuminators of opportunity” (IOs), and they can be other radars, communication systems, broadcast systems for public utility and so on. PCL radar systems have gained renewed interest over the last decades, thanks to the advantages coming from not having a dedicated transmitter, namely: (i) they can be considered as “green radars”, not introducing further electromagnetic pollution in the environment; (ii) they are generally low-cost, as they only need a receiver, whose hardware is usually easily found on the commercial market; (iii) they work in covert operation not emitting a dedicated signal, and being their position hard to be estimated. The aforementioned advantages make passive radars extremely appealing for military scenarios. To this end, in the last years, the European defence agency (EDA) funded the research project MAPIS (Multi-channel Passive ISAR for military application) aiming at investigating the target imaging and classification capabilities of passive radars based on in-

verse synthetic aperture radar (ISAR) techniques. Within this project, Fraunhofer FHR has developed a new passive radar system, SABBIA, which exploits satellite broadcasting signals like DVB-S and DVB-S2. The idea of exploiting geostationary transmissions as passive radar illuminators has been already presented in [7], [8], [9], while in [10] the application to the ISAR case has been introduced. The main advantage of a satellite-based passive radar system for ISAR is the availability of a relatively wideband signal over wide areas, also remote and/or off-shore. On the other hand, due to the relatively poor signal power level at the receiver, long integration times are needed for proper subsequent signal processing. In addition to that, high gains at the receiving antenna are also typically needed, which leads to pencil beams at the surveillance antenna. As a consequence of the aforementioned considerations, a passive radar based on geostationary illuminators is not really suited to operate in search mode. On the other hand, it might be an interesting system for non-cooperative covert target identification, which means for ISAR. In fact, in ISAR long integration times are anyway foreseen to collect data from varying target aspect angles, [11]. In addition, DVB-S2 allows for theoretically relatively wideband signals, which give the possibility to achieve range resolutions adequate for imaging purposes. Also, ISAR is usually done after a target detection stage, when the presence of a target is known together with information about its location, so that a pencil surveillance beam does not represent a major drawback. Last but not least, being DVB-S(2) a satellite based communication infrastructure, it is likely to be operative (or at least not to be easily torn down) in an actual battlefield. This represents a significant strategic advantage for military applications. The remainder of this paper is organized as follows. In Section II, the main scenarios of application are described together with their military relevance. In Section III an analysis of different possible cueing sub-systems is presented. Section IV shows some experimental setups available at Fraunhofer FHR to implement the proposed approach, and Section V shows some corresponding preliminary results. Section VI is devoted to other further potential developments, and finally in Section VII we draw our conclusions.

II. SCENARIOS AND MILITARY RELEVANCE

Let us consider the military need to perform target detection and classification in a complete silent operational mode. We propose to perform the two tasks of detection and classification using two different sub-systems. The detection task is in charge of surveying a wide volume searching for

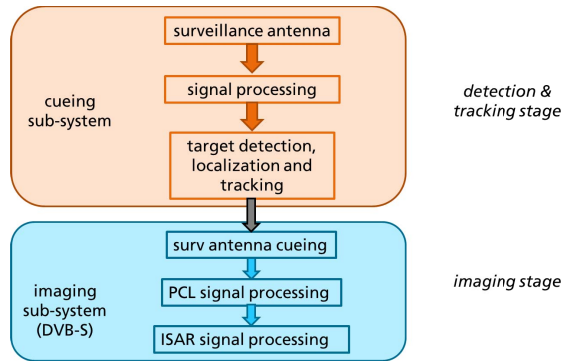


Fig. 1. Block diagram of the integrated technique

potential targets. In addition to simple detection, the target has also to be localised and continuously tracked in Cartesian coordinates. The detection stage can then be used to cue the subsequent imaging stage which finally allows target classification based on ISAR imaging. A block diagram of the resulting approach is sketched in Fig. 1. To follow this approach, the detection stage has to be silent (i.e. passive) and it has also to fulfill the subsequent requirements: (i) real-time operability: target detection, localization, and tracking must be obviously performed in real-time; (ii) localization accuracy: the target has to be localized in the 3D space with an accuracy equal or better than the beamwidth of the DVB-S antenna; (iii) reliability and availability: the target detection sub-system should operate using signal sources that are available on wide areas and possibly operative in all relevant military scenarios. The proposed approach can be potentially applied in the following military scenarios:

- *maritime scenario*: the system is located on shore. The cueing sub-system is based on passive radar exploiting either terrestrial or satellite illuminators of opportunity, while DVB-S(2) is reserved for passive target ISAR imaging. Alternatively, the cueing sub-system can be based on passive emitter tracking (PET) principle, thus exploiting the e.m. signals emitted by the target itself. A detailed analysis of pros and cons of different cueing sub-systems is provided in Sect. III. In this case, targets of interest can be naval targets approaching the coast.
- *land-based scenario*: the system is located on land. The cueing sub-system is based on passive radar exploiting either terrestrial or satellite illuminators of opportunity, while DVB-S(2) is reserved for passive target ISAR imaging. On land the availability of terrestrial illuminators like FM-radio or DVB-T is likely, although digital illuminators like DVB-T might be operated in a single frequency network (SFN) configuration, which may complicate target localization. For this scenario, a 3D localization and tracking of the targets (namely in range, azimuth, and elevation) is required, since targets of interest can be either air- or ground-borne.
- *naval scenario*: the system is located on a ship. The cueing sub-system is based on passive radar exploiting either terrestrial or satellite illuminators of opportunity,

while DVB-S(2) is reserved for passive target ISAR imaging. Alternatively, the cueing sub-system can be based on passive emitter tracking (PET) principle, thus exploiting the e.m. signals emitted by the target itself. For this scenario, a 3D localization and tracking of the targets (namely in range, azimuth, and elevation) is required, since targets of interest can be either maritime, airborne or ground-based.

III. CUEING SUB-SYSTEMS

In this Section we provide an overview of different possible cueing sub-systems. The natural choice for a silent target detection cueing sub-system is a passive radar. As is well known, passive radars may be developed upon different potential illuminators of opportunity, each of them offering peculiar characteristics. In the following sub-sections, we will mainly concentrate on terrestrial illuminators like FM-radio, and digital broadcasting communications like DAB/DVB-T. In addition, a survey over potential satellite illuminators is presented as well. Apart from passive radars, silent target detection and localization can be also performed by exploiting e.m. transmissions coming from the target itself. These techniques, referred to as passive emitter tracking (PET), are also analysed in the remainder of this Section.

A. Passive radar based on FM-radio

FM radio has been largely exploited for PCL purposes in the first years of XXIst century, given its wide availability and its reasonable transmit power. However, FM radio modulation is analog, which means that the effective instantaneous signal bandwidth is highly dependent on the program content being broadcasted. Different experimental and prototypal setups of passive radars based on FM-radio have shown detection capabilities in the medium/long range, depending on the type of targets. The limited range resolution offered by FM-radio modulation is not a critical point in the considered approach, and the low operating frequency offers also good detection against stealth targets. On the other hand, the FM-radio broadcasting infrastructure is terrestrial, and therefore it can be potentially destroyed easily in an actual battlefield scenario.

B. Passive radar based on DAB/DVB-T

DAB and DVB-T (namely terrestrial digital radio and television, respectively) use an orthogonal frequency division modulation (OFDM) scheme, which guarantees constant signal features, namely constant bandwidth and characteristics of the auto-ambiguity function. For these reasons, several experimental and prototypal passive radars are based upon such signals. Digital modulation schemes also present drawbacks mainly related to periodicities in the signal structure. These periodicities are introduced for different reasons in the transmitted signal (e.g. for synchronization purposes), but they introduce artifacts resulting in potential false alarms if not properly compensated for. In addition, DAB and DVB-T usually operate in single frequency network (SFN) scenarios, where multiple dislocated transmitters transmit the same signal at the same time. This creates issues in the association of bistatic echoes, which can be solved at the tracking stage. Experience shows that detection ranges of DAB or DVB-T based passive radars is slightly lower than with FM-radio. In addition, same issues concerning

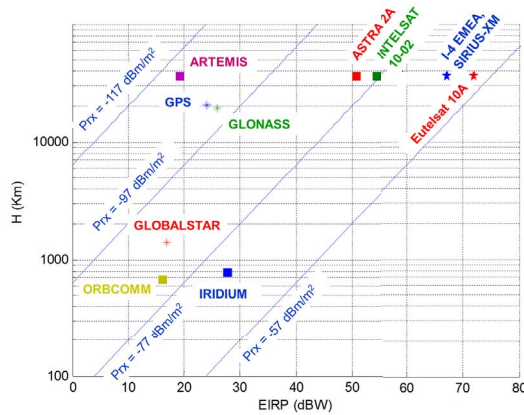


Fig. 2. Overview of space-based transmitters, from [7]

the actual availability of DAB or DVB-T infrastructure in a battlefield scenario arise.

C. Passive radar based on satellite illuminators

Satellite illuminators theoretically offer the advantage of wide coverage including remote areas of the globe, together with the intrinsic robustness against enemy tear down in a battlefield scenario. The possible use of space-based (in lieu of ground-based) illuminators, has been considered for target detection only in a limited number of studies, [7]. In fact, space-based broadcast transmitters have typical level of Equivalent Isotropic Radiated Power (EIRP) in the range of 40-50 dBW in high orbits, yielding a low level e.m. field close to the Earth's surface. An overview of possible space-based transmitters is reported in Fig. 2. When considering broadcast transmitters in Geostationary Earth Orbits (GEO), very long integration times are required to yield an acceptable Signal to Noise Ratio (SNR). This allows images to be made of the Earth's surface (also by exploiting a limited satellite motion relative to the Earth) [8] [12], but not detection of targets on mid-range using short integration times. The possibility of detecting moving targets (i.e. short integration times) can be considered using transmitters on Low Earth Orbit (LEO), [13], due to the reduced range of these sources. In this case, a full constellation (e.g. Globastar or Iridium) is necessary to guarantee a continuous coverage of a specific region and continuous satellite tracking is necessary. However, the available constellations are devoted to personal communications, so that the effective presence of signals in the air largely depends on the number of actual users and on the time-division multiplexing characteristics. Similar characteristics are offered by GNSS constellations like GPS, GLONASS, and GALILEO. In general, such potential illuminators do not guarantee an appealing solution for the proposed target detection sub-system. As an alternative, one might consider exploiting geostationary Eutelsat 10A satellite, which nominally offers sensibly higher EIRP and adequate bandwidth. This satellite has been launched in 2009 with the aim of providing broadcast satellite digital TV on mobile handheld devices (DVB-SH) in S-band. DVB-SH signals might be theoretically interesting for passive radar not only for their power level, but also for:

- Waveform characteristics (OFDM structure similar to

DVB-T);

- Area coverage (size of covered air space);
- Simplicity of operation (not requiring tracking satellite position and changing Doppler frequency).

However, over the last years the DVB-SH standard has been replaced by long term evolution (LTE) as standard for mobile reception of digital communication media. As a consequence, the actual availability of such signal in the future is uncertain.

D. Passive Emitter Tracker (PET)

PET might be an interesting solution for the cueing sub-system. It is based on the reception of e.m. signals emitted by the target itself, which are then received by multiple antennas and processed via the time difference of arrival (TDOA) principle. PET might allow target detections also in off-shore scenarios, and at medium to far ranges. On the other hand, PET requires signal transmissions from the target itself, which makes target detection impossible if the target is silent. In addition, PET provides target localisation in angle dimension only (usually in azimuth). More complex PET sub-systems might be developed for target localisation in azimuth and in elevation. Last but not least, PET sets strong hardware requirements. In fact, since the frequency transmitted by the target is unknown, the PET sub-system should listen (and digitize) over very wide frequency bands.

IV. EXPERIMENTAL SETUP AT FHR

This Section is devoted to the brief description of two components, one for target detection and cueing and one for target imaging, developed in the last years at Fraunhofer FHR. Both sub-systems are experimental setups, and therefore they offer the flexibility required to implement the proposed multi-band passive approach for target detection and imaging.

A. DVB-T PCL system ATLAS

ATLAS (see Fig. 3) is designed as a scalable, modular receiver able to realize a software defined radio based passive radar for DAB/DVB-T. This modular concept consists of four major components: (i) 12 RF receiver channel modules; (ii) 12 analogue-to-digital-converter (ADC) and 6 FPGA (field programmable gate array) modules; (iii) 12 high performance PC modules; (iv) One central local oscillator (LO) and ADC-clock generation unit. Depending on the measurement task, the system modules can be assembled such that the defined requirements are met.

1) *RF modules*: The RF modules are designed to operate at an RMS input power of up to -20 dBm. The first band filter filters the DVB-T band from 470 to 870 MHz. The signal is then amplified and mixed to the first IF (intermediate frequency) at about 1105 MHz. This intermediate frequency is optimized to allow only a small number of unwanted mixer products/images to enter up-conversion process. This second stage down-converts the signal to an IF of 80 MHz (bandwidth of 32 MHz). With an amplifier and a further band filter, the signal level is optimized for the input requirements of the ADC and adjacent spectral products are eliminated. All band filters provide sideband rejection to meet the anti-aliasing



Fig. 3. ATLAS front end

criteria for the ADC. Using two mixer stages is necessary to avoid the back-folding of adjacent channels into the desired measurement channels.

2) *ADC/FPGA modules*: The down-converted signals are digitized with a 16-bit ADC at a sampling frequency of 64 MSPS (Megasamples per second). As mentioned above the signal is placed at a center frequency f_{IF} of 80 MHz. The bandwidth is limited to 32 MHz to meet the Nyquist criterion. Sub-sampling in the 3rd Nyquist band eliminates unwanted coupling of power-supply noise into the signal path, as power-supply noise is usually limited from DC to 5 MHz. The FPGA preprocesses the incoming data from up to two channels and transfers them to the high-performance PCs via PCI Express. The FPGA firmware provides several trigger scenarios (internal, external pulsed/gated, threshold) and recording modes (continuous, triggered, fixed length). The data downstream is realized via PCI Express x8 Gen1. Using multistage buffering and SG-DMA (scatter gather-direct memory access) methods, data rates of 1.2 GByte/s can be achieved (real measurement).

3) *High-performance PC modules*: The post-processing is implemented on state of the art high-performance PC server modules. Each module processes 2 channels. The modules provide 40 GBit/s Infiniband networking interfaces for high-speed module-to-module communication, which is recommended e.g. for beam forming. A RAID-0 HDD array enables the continuous storage of data up to 300 MByte/s. The needed processing power can also be scaled down to reduce costs, as real-time requirements may vary depending on the measurement task.

B. DVB-S(2) PCL system SABBIA

The SABBIA system has been developed by FHR within the MAPIS project. It is designed to generate ISAR images of cooperative and non-cooperative targets (both airborne and maritime) exploiting broadcast satellite DVB-S(2) signals emitted from geostationary satellites, such as ASTRA 19.2E. The SABBIA system is based on two identical receiver frontends, one for surveillance and the other for reference. The surveillance channel tracks the target (e.g. airplanes or ships) while the reference channel points to the geostationary satellite which illuminates the scene. The signals from both frontends are recorded after analog-to-digital conversion with high-speed



Fig. 4. SABBIA reference (left) and surveillance (right) antennas

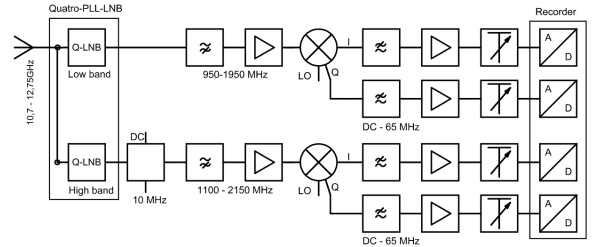


Fig. 5. SABBIA frontend receiver block diagram

data recorders. The antenna units consist of a custom designed antenna horn, a Quad-LNB (low-noise-block) with an external 10 MHz signal reference and low phase-noise oscillator, an 85 cm dish and a tripod. The antennas are connected to the RF receiver. Each antenna is also connected to a GPS/IMU unit to obtain precise information about location and antenna pointing direction. Fig. 4 shows the two SABBIA antennas. On each antenna, the IMU/GPS unit can be seen on top of the dish, while the Quad-LNB is located right behind the antenna feed, with the 4 cables corresponding to H- and V- polarisations on low- and high- bands. The reference antenna is also equipped with an optical camera, which helps in the identification of the antenna pointing direction (see also Fig. 8). The Quad-LNB converts the high- and low-bands of the DVB-S signal, each with vertical and horizontal polarizations simultaneously. This potentially enables a full polarimetric operation. However, SABBIA is developed to be extremely flexible, so that polarimetric operation, if not needed, can be swapped for higher digitized signal bandwidth or for other characteristics. The 10 MHz reference ensures a coherent operation of the LNB with the rest of the system. A detailed block diagram of the frontend receiver is reported in Fig. 5. Here one can see that the DVB-S signal impinges the antenna at radio frequency between 10.7 GHz and 12.75 GHz. The expected level of the (reference) signal is pretty low, approximately -125 dBm at antenna level. The Quad-LNB amplifies the signal and down-converts it to 950-1950 MHz (DVB-S low-band) and 1100-2150 MHz (DVB-S high-band), respectively. After that, additional filtering and amplification is applied. Then the signal is further down-converted into I/Q-signals. The IF-bandwidth of the I/Q-signals is currently set to 32 MHz. A combination of amplifiers and digitally adjustable attenuators ensures an ideal levelling of the signal prior than digital sampling. A picture of the receiver frontend is shown in Fig. 6. The ADC then samples the signals with 64 MSPS each. The clocks and local

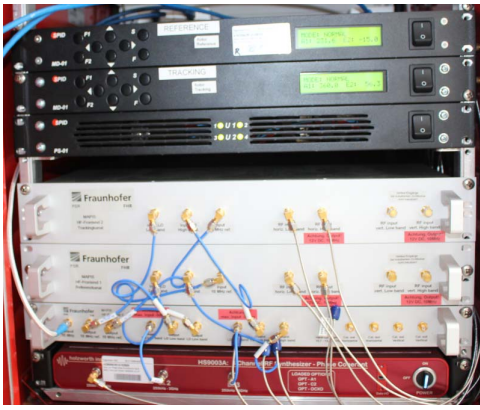


Fig. 6. SABBIA receiver frontend



Fig. 7. SABBIA ADCs and data recording

oscillators are generated by an ultra-low phase-noise DDS based synthesizer. The ADCs and the data recording units are shown in Fig. 7. The flexibility of the SABBIA system allows for future increase of the digitized signal bandwidth, in order to fulfill the resolution requirements for ISAR imaging. The system is controlled by an instrument control software (ICS) internally developed at FHR. ICS allows the steering of both antennas either manually or automatically, the adjustment of the attenuators, the swap of receiving channels etc. Fig. 8 shows the ICS graphical interface with an live tracking image of different setting options and signal monitoring.

V. RESULTS

A first deployment of the proposed multi-band approach has been conducted within the MAPIS field trial which took place in Livorno, Italy, in June 2017. Both aforementioned systems were installed on the roof of the Istituto Vallauri at the Italian Naval Academy located directly on the coast line in front of the Tyrrhenian Sea (see Fig. 9). The exploited transmitters of opportunity were a DVB-T transmitter located at Monte Serra in Pisa (around 33 km North-East from the receiver, Fig. 10) and the ASTRA satellite, respectively. The result of the PCL processing by the cueing DVB-T based sub-system is shown in Fig. 11. In this case, a track has been

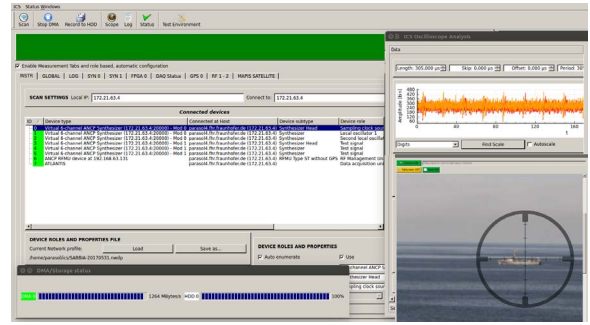


Fig. 8. SABBIA instrument control software (ICS). The window on the bottom right is the camera, which shows the correct pointing of the surveillance antenna towards the target



Fig. 9. SABBIA System on the roof of the Istituto Vallauri in Livorno

established from detections of the cooperative military ship Porpora acquired during the third day of measurements. As one can see, the target is located in Cartesian coordinates so that the target position information can be used to cue the SABBIA sub-system. Fig. 12 reports the ISAR image obtained with the SABBIA sub-system, that is exploiting the DVB-S2 satellite signal. More detailed description of the ISAR results can be found in [4].

VI. FURTHER DEVELOPMENTS

The installation of the proposed system on a moving platform (such as a boat or ship) sets additional challenges to



Fig. 10. Relative geometry between DVB-T transmitter in Monte Serra and receiving site at the Italian Naval Academy

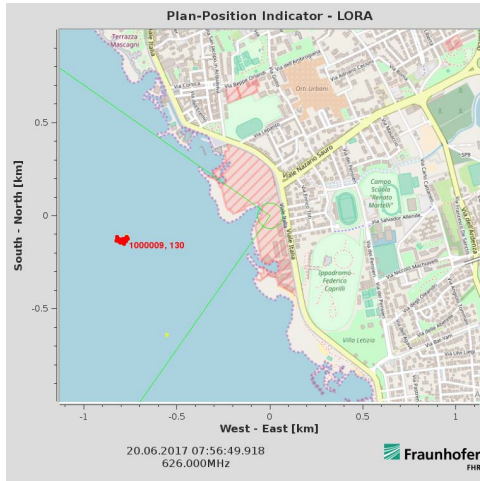


Fig. 11. Established track of the cooperative target Porpora obtained with ATLAS

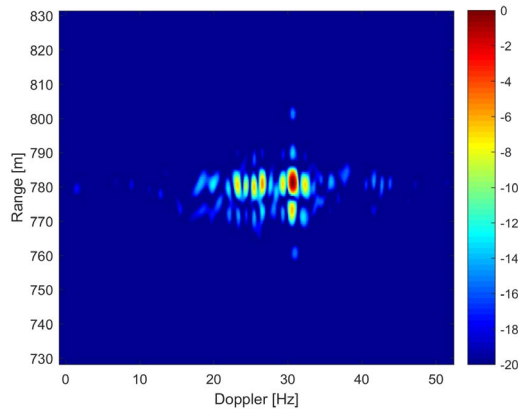


Fig. 12. ISAR image of the ship Porpora obtained with the imaging sub-system SABBIA, [dB]

the target detection/cueing sub-system, if passive radar is used for this purpose. In fact, the motion of the receiving platform might require extra processing to ensure reliable target detection. Different approaches have been recently proposed for platform motion compensation in passive radars using single surveillance channel ([14]) and multiple channels (namely leading to displaced phased center antenna, DPCA, and space-time adaptive processing, STAP, for passive radars, [15] [16] [17]).

VII. CONCLUSIONS

In this paper, a novel approach for silent target detection and imaging is proposed. The approach is based on two passive sub-systems, the detection one and the imaging one. The detection sub-system should cue the imaging one, giving information about the presence and the location of the target. Different possible solutions for the cueing sub-system are identified and discussed. The imaging sub-system is based on passive ISAR exploiting satellite illumination (namely DVB-S(2)). An existing hardware setup available at Fraunhofer FHR and able to implement the proposed approach is described.

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REFERENCES

- [1] D. Gromek, K. Kulpa, and P. Samczynski, "Experimental results of passive SAR imaging using DVB-T illuminators of opportunity," *IEEE Geoscience and Remote Sensing Letters*, vol. 13, pp. 1124–1128, Aug 2016.
- [2] J. Garry, G. Smith, and C. Baker, "Wideband DVT passive ISAR system design," in *IEEE Intl. Radar 2015*, 2015.
- [3] F. Berizzi and et al, "Multichannel passive ISAR imaging for military applications (MAPIS)," in *Specialist Meeting NATO SET-231 on Multi-Band Multi-Mode Radar*, 2016.
- [4] I. Pisciotto, D. Cristallini, J. Schell, and V. Seidel, "Passive ISAR for maritime target imaging: Experimental results," in *Proceedings of IRS 2018*, 2018.
- [5] A. Manno-Kovacs, "Automatic target classification in passive ISAR range-crossrange images," in *IEEE Radarconf 2018*, 2018.
- [6] P. Howland, "Editorial: Passive radar systems," *IEE Proceedings - Radar, Sonar and Navigation*, vol. 152, pp. 105–106, June 2005.
- [7] D. Cristallini, M. Caruso, P. Falcone, D. Langellotti, C. Bongioanni, F. Colone, S. Scafe, and P. Lombardo, "Space-based passive radar enabled by the new generation of geostationary broadcast satellites," in *IEEE Aerospace Conference*, 2010, pp. 1–11, March 2010.
- [8] C. Prati, F. Rocca, D. Giancola, and A. M. Guarnieri, "Passive geosynchronous SAR system reusing backscattered digital audio broadcasting signals," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 36, pp. 1973–1976, Nov 1998.
- [9] M. Antoniou and M. Cherniakov, "GNSS-based bistatic SAR: a signal processing view," *EURASIP Journal on Advances in Signal Processing*, vol. 2013, no. 1, p. 98, 2013.
- [10] D. Pastina, M. Sedehi, and D. Cristallini, "Passive bistatic ISAR based on geostationary satellites for coastal surveillance," in *IEEE Radar Conference*, 2010, pp. 865–870, May 2010.
- [11] V. C. Chen and M. Martorella, *Inverse Synthetic Aperture Radar Imaging: Principles, Algorithms and Applications*. Radar, Sonar & Navigation, Institution of Engineering and Technology, 2014.
- [12] L. Cazzani, C. Colesanti, D. Leva, G. Nesti, C. Prati, F. Rocca, and D. Tarchi, "A ground based parasitic SAR experiment," in *Geoscience and Remote Sensing Symposium, 1999. IGARSS '99 Proceedings. IEEE 1999 International*, vol. 3, pp. 1525–1527 vol.3, 1999.
- [13] R. Saini and M. Cherniakov, "DTV signal ambiguity function analysis for radar application," *IEE Proceedings - Radar, Sonar and Navigation*, vol. 152, pp. 133–142, June 2005.
- [14] J. Palmer, M. Ummenhofer, A. Summers, G. Bournaka, S. Palumbo, and D. Cristallini, "Receiver platform motion compensation in passive radar," *IET Radar, Sonar Navigation*, vol. 11, no. 6, pp. 922–931, 2017.
- [15] B. Dawidowicz, K. Kulpa, M. Malanowski, J. Misiurewicz, P. Samczynski, and M. Smolarczyk, "DPCA detection of moving targets in airborne passive radar," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 48, pp. 1347–1357, APRIL 2012.
- [16] P. Wojacek, F. Colone, D. Cristallini, P. Lombardo, and H. Kuschel, "The application of the reciprocal filter and DPCA for GMTI in DVB-T PCL," in *IET RADAR 2017*, 2017.
- [17] P. Wojacek, F. Colone, D. Cristallini, and P. Lombardo, "Reciprocal filter-based STAP for passive radar on moving platforms," *submitted to IEEE TAES*, 2018.